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Evaluation of different air dosing strategies to enhance H_2S removal in microaerobic systems treating low-strength wastewaters

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Abstract: This study aimed to evaluate different air dosing strategies such as microaeration flow rates and air dosing points to enhance H_2S removal in microaerobic systems treating low-strength wastewaters. Efficiency and stability of the reactors as well as biogas quality were assessed, and microbial community changes were evaluated using PCR-DGGE technique. The results showed that the air dosing point affected the H_2S concentration and that air dosing at the headspace promoted the highest H_2S removal efficiency. The airflow rate also affected the process, since H_2S concentration in the biogas was higher at 0.1 mL air.min^{-1} than at 0.3 mL air.min^{-1}. The methane concentration in the biogas was also affected by both air dosing point and flow rate, since the lowest value was observed at the highest airflow rate of the headspace dosing point, due to dilution by the N_2 influx applied to the system. The highest productivity and operational efficiency were observed at this air dosing point, with this airflow ($HD_{0.3}$), which corroborates with the operational results and the ecological parameters, since the microaeration at this stage promoted high bacterial and archaeal species

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richness and diversity, optimum functional organization, high COD and H_2S removal efficiencies.

Keywords: anaerobic treatment; biogas; hydrogen sulfide; microaeration; odor control.

1. Introduction

Anaerobic wastewater treatment is a well-established technology in which conversion of complex organic matter into biogas is carried out by different microbial species from the Bacteria and Archaea domains [1]. The composition of biogas depends on the type of carbon source supplemented and the operational conditions of the reactor [2]. Although H_2S is usually found in very low concentration in the biogas compared to CH_4 and CO_2, this compound is responsible of some significant problems in wastewater treatment plants due to its odorous and corrosive nature, which is highly undesirable in energy recovery processes [3]. Furthermore, H_2S reacts with metals [4], corroding iron, copper, cement etc. [2]. Thus, in order to use biogas for energy recovery purposes, cost-effective and sustainable methods are needed for sulfide removal.

Available technologies such as controlling H_2S emissions by adding sulfidogenesis inhibitors, adsorption on activated carbon [1], gas washing [4] and other available techniques [5], generally imply high investment and/or operational costs. In this context, removing H_2S from biogas through microaerobic processes seems to be an efficient and sustainable alternative to the high costs and high environmental impact of chemical adsorption and precipitation of sulfides [6, 7, 8, 9]. The microaerobic biotechnology consists on applying a limited influx of oxygen (O_2) into an anaerobic bioreactor. This strategy avoids the need for additional units for biological H_2S oxidation, since sulfide-oxidizing bacteria, which promote H_2S oxidation, can be found in the inoculated sludge [8]. Furthermore, moderate aeration in anaerobic reactors treating wastewaters with high sulfate and/or sulfide concentrations reduces H_2S in biogas and thus sulfide toxicity to methanogenic archaea. This occurs due to the fact that H_2S oxidation surpasses other oxidative processes [10].

Most of the studies using the microaerobic technology have been performed for eliminating H_2S during sludge digestion and during the treatment of sulfate-rich wastewaters. These studies have already confirm the potential of microaeration for H_2S removal without promoting negative impacts on anaerobic digestion [9, 10, 11, 12].

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However, fewer studies have addressed the potential of this technology for the treatment of low-strength wastewaters (low COD and sulfate concentrations). Moreover, further investigation is required concerning the process engineering regardless of the type of wastewater in order optimize the technology. Key aspects that need to be addressed include: the best point for air dosing, the most adequate airflow rate, proper diffusion devices, process control, the microbiology of the process, the robustness of the technology towards operational failures or fluctuations (shocks loads, interruptions of aeration etc.), the improvement of the system capacity for dissolved sulfide removal etc.

Changes of microbial communities (microbial succession) engineered ecosystems (bioreactors) in response to variations in operating conditions constitute a main issue in microbial ecology to predict system or ecosystem behavior [13]. Several characteristics of the microbial community structure seem to play a key role in maintaining functional stability under these variations. For example, richness, evenness, dynamics, functional redundancy, microbial composition and microbial interactions seem to be of utmost importance for controlling reactor and ecosystem functioning [14, 15, 16].

As an example, communities exhibiting intermediate evenness values (some species are dominant but most of them are present in decreasing lower amounts) have been shown to better deal with changing environmental conditions, since they have a pool of less dominant species able to replace the leading ones under operating or environmental fluctuations [17]. Species richness effects on ecosystem functioning can decrease when functionally redundant species exist in the community [14].

This type of studies helps to understand process performance and behavior not only in anaerobic but in every biological ecosystem towards fluctuations in environmental conditions. Moreover, these studies help to discover general trends that can be used to optimize and/or predict the behavior of bioreactors.

In this context, two microaerobic reactors treating low-strength wastewaters were operated in this study aimed at: 1) evaluate the best point for air dosing at different zones of the reactor; 2) assess the effects of increasing the airflow rate on H_2S and dissolved sulfide removal efficiencies; 3) evaluate the operational stability of the reactors; 4) assess the quality of the biogas produced. A sulfur balance analysis was calculated and bacterial and archaeal communities were studied by means of molecular biology techniques for a better understanding of the process and assess the effect of oxygen on the anaerobic ecosystem.

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2. Material and methods

2.1. Inoculum and carbon source

The reactors were inoculated with granular anaerobic sludge according to Sousa, Oliveira [9].

The feeding consisted of synthetic wastewater with ethanol as the electron donor (COD = 1 $g.L^{-1}$) (99.8% purity, Dinâmica Química, Brazil). According to literature, defined substrates, such as lactate, ethanol, methanol and hydrogen, are preferential electron donors for sulfate reduction, in which high efficiencies are obtained (~ 80%) [18, 19]. The composition of basal medium was according to Firmino, Silva [20]. To increase the sulfide concentration in the biogas, sodium sulfate (400 $mg \cdot L^{-1}$) was added to the basal medium. The influent was stored under refrigeration at a temperature of approximately 4 °C. To keep pH around 7.0, the wastewater was buffered with sodium bicarbonate ($NaHCO_3$) in the proportion of 1 g $NaHCO_3$ to each 1 g COD ethanol.

2.3. Reactors set up

The experiment was conducted in two lab-scale continuous-flow anaerobic reactors (R1 and R2) with a total volume of 3.5 L each one (working volume of 2.8 L, digestion compartment inner diameter of 60 mm, settling compartment inner diameter of 85 mm, total height of 820 mm). The reactors were made of acrylic plastic and operated at a hydraulic retention time (HRT) of approximately 11 h. An organic loading rate (OLR) of about 2.0 kg $COD \cdot m^{-3} \cdot d^{-1}$ was applied. The reactors were fed by a peristaltic pump (Watson Marlon, Model 323U/D) at an average flow rate of 6.0 $L \cdot d^{-1}$. The reactors effluent were recirculated by metering pumps (ProMinent, model Concept Plus CNPA 1000 NPB2 00A01) at a flow rate of approximately 1 $L \cdot h^{-1}$, which provided an upflow velocity of 0.5 $m \cdot h^{-1}$. The systems were equipped with electrodes for on-line redox potential (ORP) measurements. Biogas production was measured by counting the number of pulses generated in a water displacement device (25 mL per pulse).

2.4. Operational strategy

The R1 system was initially operated under anaerobic conditions (AN) for 155 days. After that, microaeration was applied above the sludge blanket (ASB) in two different airflow rates of 0.1 and 0.3 mL min^{-1}, for 90 and 64 days, respectively. The R2 system was operated under microaerobic conditions during the whole experiment. Two

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microaeration application points (stages) were studied, at the headspace (HD) and directly at the sludge blanket (SB) with the same airflow rates (0.1 and 0.3 mL min^{-1}). The details about the operational parameters of reactors R1 and R2 during the entire experiment are summarized in Table 1.

2.5. Analytical methods

Chemical oxygen demand (COD), sulfate and dissolved sulfide were analyzed according to APHA [21], whereas biogas characterization was performed according to Sousa, Oliveira [9]. Bacterial and archaeal community structure was analyzed by PCR-DGGE (Ploymerase Chain Reaction Denaturing Gradient Gel Electrophoresis) according to [9]. The obtained archaeal and bacterial DGGE gels were analyzed in order to calculate the following ecological parameters: evenness/functional organization (*Fo*), ranged weighted richness (*Rr*) and Shannon diversity index (*H*) according to [22].

2.6. Statistical analysis

The experimental data sets of the different periods were compared to each other by the Mann-Whitney Rank Sum test at a 95.0% confidence level [9].

3. Results and discussion

3.1. Effect of different air dosing strategies on COD removal and operational stability

In the anaerobic phase (AN) of R1, the average COD removal efficiency was 80% (Table 2). Subsequently, after the microaeration was applied above the sludge blanket (ASB), the average values of COD removal were higher than in the anaerobic phase: 92% ($ASB_{0.1}$) and 87% ($ASB_{0.3}$) (Table 2). However, there was no statistically significant difference between the microaerobic periods using different airflow rates ($ASB_{0.1}$, $ASB_{0.3}$) ($p = 0.46$). Furthermore, there was no statistically significant difference between the stages AN and $ASB_{0.1}$ ($p = 0.33$), and between stages AN and $ASB_{0.3}$ ($p = 0.32$).

In R2, COD removal efficiencies, when microaeration was applied at the headspace, were 89%, using an airflow rate of 0.3 mL min^{-1} ($HD_{0.3}$), and 84%, using an airflow rate of 0.1 mL min^{-1} ($HD_{0.1}$). However, no statistically significant differences were observed between these values ($p = 0.50$). When the microaeration was applied at

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the sludge blanket (SB), COD removal efficiencies of 79 ($SB_{0.1}$) and 84% ($SB_{0.3}$) were recorded, presenting again no statistically significant difference between them ($p = 0.67$).

In this work, the microaeration either applied into the liquid or gaseous phase did not show any significant effect on the COD removal capacity of the systems. These results are in accordance with Krayzelova, Bartacek [23], who evaluated the microaeration at the sludge blanket and found COD removals of 89 and 90%, for the anaerobic and microaerobic reactors, respectively. They also did not observe a negative effect on the methanogenic activity of the microbiota.

Regarding operational stability, the reactor was stable during the whole experiment, and no pH change (~7.5), VFA accumulation (VFA < 250 $mg \cdot L^{-1}$) or biomass loss were observed. These results are consistent with those reported in literature, since it has been widely observed that the influx of small amounts of O_2 into anaerobic digesters do not affect negatively the digestion performance [24, 25]. In fact, Rodríguez, Lopes [26] and Zhou, Imai [27] found a methanogenic activity decrease due to the presence of high sulfide concentrations under anaerobic conditions, and the microaeration was a good strategy to recover the activity by oxidizing part of the sulfide. Rodríguez, Lopes [26] used COD/SO_4^{-2} ratios of 2-4 and sulfate concentrations of 1440 and 2880 $mg \cdot L^{-1}$. Zhou, Imai [27] used COD/SO_4^{-2} ratios of 15-20 and sulfate concentrations ranging from 300 to 500 $mg \cdot L^{-1}$. The COD/SO_4^{-2} ratio of the current experiment was 7, and the sulfate concentration was about 140 $mg \cdot L^{-1}$.

The fact that the reactors in this work were inoculated with granular sludge may have contributed to the maintenance of COD and sulfate removal performance, as well as methane production, when very low O_2 concentrations were applied. Shen and Guiot [28] mentioned that methanogenic microorganisms in granular sludge are more tolerant to oxygen than in flocculent sludge. According to those authors, such a behavior is due to the presence of a multilayer structure in granular sludge, in which the facultative microorganisms are predominant in the outer part of the granule, and the methanogenic archaea, more sensible to oxygen, would be protected in the granule core.

On the other hand, a detrimental effect was not found when the air dosing was applied into the liquid phase (ASB and SB) in this work. Likely, the sulfide formed from sulfate reduction was not at a concentration, which could cause a toxic effect on methanogens. According to the presented results (Table 2), sulfide concentration at all stages was lower than 50 $mg \cdot L^{-1}$. Lettinga, Pol [29] reported inhibition from 25 to 90%

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of the methanogenic activity for a sulfide concentration of 150 $mg \cdot L^{-1}$. Karhadkar, Audic [30] found a 50% inhibition of the methanogenic activity for sulfide concentrations of 100-224 $mg \cdot L^{-1}$.

H_2S removal

The operational performance for the different air dosing strategies is shown in Table 2. The biogas produced at the anaerobic stage (AN) of R1 contained 0.5% of H_2S. The airflow applied above the sludge blanket (ASB) at stage $ASB_{0.1}$ promoted a decrease of sulfide concentration to 0.34%. When the airflow rate was increased, during stage $ASB_{0.3}$, sulfide concentration decreased further to 0.17%.

The microaeration applied directly at the headspace (HD) in R2 also had a positive impact on H_2S removal, in which the concentrations were 0.07% and 0.03%, respectively, for the stages $HD_{0.1}$ and $HD_{0.3}$.

The last microaeration strategy evaluated was directly at the sludge blanket (SB) in R2, based on the hypothesis that the flow could strip the dissolved H_2S to higher zones of the reactor, including the headspace, thus favoring a more effective oxidation of sulfide. However, at both stages $SB_{0.3}$ and $SB_{0.1}$, the concentration of H_2S increased up to 4.6 times compared to the level obtained at stage $HD_{0.3}$, i.e. increasing from 0.03 to 0.10% ($SB_{0.1}$) and 0.13% ($SB_{0.3}$). Therefore, applying air at the sludge blanket zone increased a bit the sulfide stripping from the liquid phase but it did not promote an efficient H_2S removal from the biogas.

Some investigations report sulfide reductions of 20-30% in the liquid phase and high H_2S removal from the biogas [23, 27, 31]. It is worth mentioning that most of sulfide oxidation was probably taking place at the headspace, even when the air dosing was in the liquid phase.

Krayzelova, Bartacek [23] found that the average concentration of H_2S in the biogas from a UASB reactor was 8.9 and 2.2 $g \cdot m^{-3}$ when operated under anaerobic and microaerobic conditions, respectively, representing an improvement of 73% compared to the anaerobic phase. They applied microaeration at the feeding line, below the sludge blanket, at a flow rate of 0.7 $mL \cdot min^{-1}$ air (0.5 mol $O_2 \cdot mol^{-1}$ S^{-2}). These results are in accordance with the present investigation, since an improvement of 80% and 74% of H_2S removal was obtained in phases $SB_{0.1}$ and $SB_{0.3}$, respectively, when compared to the anaerobic conditions. In the present work, the airflow rates used were 0.1 and 0.3 $mL \cdot min^{-1}$, corresponding to 0.15 and 0.45 mol $O_2 \cdot mol^{-1}$ S^{-2}, respectively.

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Moreover, the efficiency of desulfurization was not as high as in other studies [24, 32], in which a removal efficiency of 98-99% was observed. However, the dose of air or oxygen was much higher in those studies compared to present study.

When comparing the air dosing strategies, it can be noticed that the microaeration at the headspace was the most efficient method for H_2S removal in low-strength wastewaters. These results are similar to those observed by Díaz, Pérez [32] by evaluating the microaerobic process in a sludge digester. The latter work assessed different air dosing points, and the headspace presented the best performance. Previous microbial community analyses have revealed that the sulfide oxidizing bacteria (SOB) were growing mainly on the headspace surface wall or on the gas-liquid interface, suggesting that the biological sulfide oxidation was occurring at this point [26, 32, 33, 34].

SO_4^{-2} removal

Under anaerobic conditions (AN), the average removal efficiency reached 83.4% (Table 2). Microaeration above the sludge blanket (ASB) during stage $ASB_{0.1}$ promoted an increase in sulfate removal efficiency to 86%. However, at stage $ASB_{0.3}$, the efficiency decreased to 77% (Table 2). A statistically significant difference ($p = 0.03$) was observed between the microaerobic stages $ASB_{0.1}$ and $ASB_{0.3}$. The stage $ASB_{0.3}$ was also slightly less stable in terms of sulfate reduction, probably due to the higher concentration of O_2, which might have favored the re-oxidation of sulfide or elemental sulfur to sulfate. However, this hypothesis is very unlikely to occur because the molar flow rate (mmol $O_2 \cdot mmol^{-1}$ S^{-2}) applied was lower than 0.5. It is reported that, only when the molar flow rate is higher than 2.0, sulfide is completely oxidized to sulfate [35].

Microaeration applied directly at the headspace (HD) provided sulfate reduction efficiency of 78% and 81%, at stages $HD_{0.1}$ and $HD_{0.3}$, respectively, which were statistically similar ($p = 0.37$) (Table 2). Therefore, apparently the presence of microaeration at the headspace and the increase in air dosage did not affect significantly the system sulfate removal.

The levels of sulfate reduction, when the microaeration was applied at the sludge blanket point (SB), were 79% and 68%, at stages $SB_{0.1}$ and $SB_{0.3}$, respectively (Table 2). A statistically significant difference ($p = 0.01$) was observed between the airflow rates.

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Dissolved sulfide removal

The mean value of dissolved sulfide concentration at the anaerobic stage (R1) was 30 $mg \cdot L^{-1}$ (Table 2). Microaeration above the sludge blanket (ASB) at stages $ASB_{0.1}$ and $ASB_{0.3}$ provided concentrations of 29 and 26 $mg \cdot L^{-1}$, respectively, which were not statistically different ($p = 0.65$). Concerning microaeration at the headspace (R2), no significant difference ($p = 0.61$) was observed between the stages $HD_{0.1}$ and $HD_{0.3}$, reaching concentrations of dissolved sulfide of 25 and 24 $mg \cdot L^{-1}$, respectively. The sulfide concentration at stages $SB_{0.1}$ and $SB_{0.3}$ were identical (24 $mg \cdot L^{-1}$) ($p = 0.73$).

According to the results, for all air dosing points tested, the increase in the air dosage did not have any significant effect on sulfide removal from the liquid phase. Moreover, when microaeration was applied above the sludge blanket, sulfide concentration increased compared to the anaerobic phase. However, for microaeration at either the headspace or the sludge blanket, just a small fraction of the dissolved sulfide was removed (16%).

Dosing air into the liquid phase also caused a decrease in sulfide concentration in the liquid phase, usually about 20-30% [23, 27, 31, 32]. Díaz, Pérez [32] reported that, when the microaeration was at the headspace, only H_2S was possible to be removed, and the dissolved sulfide was only removed when microaeration was dosed in the liquid phase.

In the present study, it was possible to remove the dissolved sulfide even when the microaeration was at the headspace. As mentioned before, the decrease of dissolved sulfide concentration in the liquid phase has the additional positive effect of decreasing sulfide toxicity towards methanogens and helping odor control in the treatment plant because there is less gas striping in the effluent.

Sulfur mass balance

A sulfur mass balance was done in order to understand the main flows of sulfur conversions. In general, three sulfur flows were considered: sulfur load in terms of sulfate ($S\text{-}SO_4^{-2}$) and sulfide ($S\text{-}S^{2-}$) of influent and effluent, as well as hydrogen sulfide in the biogas ($S\text{-}H_2S$). The elemental sulfur was not considered due to the difficulty of measuring it, since it may be adhered to the tubing walls or to the reactor headspace inner walls, adsorbed onto the sludge or even as particulate matter in the effluent. Sulfur mass balance for the different air dosing strategies is shown in Fig. 1.

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It can be noticed that most of the sulfur remained in the liquid phase in the form of sulfate (about 20%) and dissolved sulfide (about 65%). The H_2S formed, even during the anaerobic phase, only represented a very small fraction of the total sulfur (less than 5%). The gap verified in almost all cases (28-4%) is attributed to the sulfur incorporated into the biomass, precipitation of sulfide by some metals present in the micronutrients solution, as well as undetected products such as thiosulfate. These findings for gaps in mass balance are also reported elsewhere [23, 32, 36].

When oxygen is present during the microaerobic process, other factors interfere in mass balance, such as formation of elemental sulfur, which can be retained in the sludge blanket or even at the headspace of the reactor. In the present experiment, a sulfuric mass (similar to elemental sulfur) was found adhered to the inner walls of the headspace, a common occurrence, as observed by other authors [23, 32, 36].

The elemental sulfur deposition on the headspace wall and tubes may obstruct the system, resulting in an overpressure and a biogas leak [26, 32, 33, 34]. However, Jeníček, Horejš [7], while monitoring a real scale digester with microaeration at sludge recirculation line, did not observe any clogging of the biogas line. According to the latter investigation, such a behavior was due to the controlled air dosing in the liquid phase, in which most of the oxygen was consumed in the digester.

During the anaerobic step, 17% of the effluent sulfur was in form of sulfate, and 62% as dissolved sulfide (Fig. 1). When microaeration was supplemented at the reactor headspace ($HD_{0.3}$), the percentage of dissolved sulfide was reduced to 53%. These results are in accordance to the ones reported by Krayzelova, Bartacek [23], who found that the sulfide detected in the effluent from microaerobic and anaerobic reactors accounted for 41% and 49% of the sulfur dosed, respectively. These authors highlighted that the main differences between the anaerobic and microaerobic reactors were the amount of hydrogen sulfide in biogas, the deposition of sulfur in the G–L–S separator, and solids in the effluent excluding sulfide. According to them, over 33% of sulfur was present in the solid phase of the effluent from the microaerobic reactor, while only 6% was in the effluent of the anaerobic reactor. Additionally, large part of the elemental sulfur oxidized during microaeration also accumulated in the G–S–L separator of the microaerobic reactor (10%), even when the microaeration was applied at the bottom of the reactor [23].

One important point to be observed is that, although the microaeration was very effective to remove H_2S, being an *in situ* strategy for biogas treatment, the effect on

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dissolved sulfide was not evident, and this fraction showed to be the most important one in the effluent mass balance. Therefore, strategies to combine simultaneous H_2S and dissolved sulfide are very important in order to control odor in treatment plants and produce a cleaner biogas.

3.2. Effect of different air dosing strategies on biogas quality

Average values of concentration (%) and molar flow rate ($mmol \cdot d^{-1}$) for the gases O_2, N_2, CH_4, CO_2, H_2S and NH_3 are shown in Table 3.

Methane molar flow rate was higher during the anaerobic stage compared to all other stages when microaeration was applied (Table 3). Another aspect is that, when the microaeration was higher, methane molar production was lower. Most likely, part of the organic matter was oxidized by facultative microorganisms present in the anaerobic sludge, which prefer oxygen as electron acceptor, therefore decreasing the organic fraction available for anaerobic microorganisms to produce methane.

Literature also reports a possible use of methane as electron donor when oxygen is present, however this is very unlikely to occur in the conditions tested and the source of sludge used. According to Ramos, Pérez [37], eventual increases in aerobic activity inside the reactor could also explain occasional decreases in methane concentration on the biogas.

As expected, microaeration dilutes the methane in the biogas and increases the concentration of oxygen and nitrogen (Table 3). The latter increase is because atmospheric air is used as source of oxygen. However, in all cases, the concentration of methane is much higher than 40%, which is the limit value for the combustion of methane.

In general, for safety reasons, mixing oxygen or air with biogas is not allowed due to the explosion risks of the mix methane/oxygen. However, the inflammable range is typically from 5 to 15% methane:air (v:v) [4]. Therefore, the leak of methane in the air should be considered a higher risk compared to the mix of a small amount of oxygen/air with biogas.

According to Jeníček, Horejš [7], during the usual microaerobic desulfurization, the amount of oxygen in the biogas very unlikely will reach those values. The maximum concentration of O_2 in the biogas was 0.3% during a full-scale investigation with microaerated anaerobic digesters. In the current investigation, the concentration of O_2 in

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the biogas was by far lower than the threshold limit, which is in agreement with the latter investigation.

CO_2 concentration values in biogas are lower than typical values found in biogas (Table 3). However, the percentage of CH_4 and CO_2 in biogas largely depends on the feedstock, and on the experimental conditions applied. In this work, ethanol was used as the sole carbon and energy source, and its fermentation does not generate directly CO_2, but it does H_2 and acetate, which are methanogenic substrates. In fact, CH_4 and CO_2 can be produced from acetoclastic methanogenesis. However, this CO_2 can be used as electron acceptor by hydrogenotrophic archaea to produce CH_4 from H_2. Therefore, it might have reduced CO_2 concentration in biogas, although further investigation is necessary to reinforce this hypothesis. Additionally, microaeration also diluted the CO_2 concentration in the biogas, as reported for methane.

NH_3 molar flow rate was higher during the anaerobic stage compared to all other stages when microaeration was applied (Table 3). Probably, a biological reaction was taking place, especially when the microaeration was applied to the headspace. However, further investigations need to be conducted in order to elucidate if the removal is via assimilatory or dissimilatory process.

3.3. Microbial community structure under different air dosing points and airflow rates

Changes in operational conditions in bioreactors promote changes in microbial interactions, leading to the adaptation of the species to the environment imposed. In this context, analyzing changes in microbial community structure (Fig. 2) in terms of different ecological parameters such as richness, evenness, diversity, etc. during reactor operation, have been widely used to understand and trying to predict the behavior of the system [17]. Various authors have shown that even a strictly direct correlation between the functionality of a biological system and the ecological parameters observed are not observed, maintaining certain ecological parameters within certain ranges promotes stability of bioreactor performance [15, 38].

In this study, bacterial and archaeal communities in the seed sample showed richness, functional organization (Fo) and Shannon diversity values within the range that, according to previous studies, favor the stability of the reactor due to a rapid adaptation of the community [17]. High richness, and diversity as well as a medium functional organization were observed (Table 4). In this context, despite the inoculum presented optimum characteristics in terms of community structure, a clear impact of

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ethanol introduction was observed in the bacterial and archaeal communities of R1 during stage AN (Table 4). While the bacterial community was slightly affected (a reduction in range-weighted richness (Rr) was observed but still remained within the range of medium Rr values), the archaeal community suffered a major impact; Fo decreased from medium to low values, and Rr decreased from high to low values, with the concomitant reduction of H (which includes evenness and richness parameters on its calculation). Bacteria present a more versatile metabolism and the community was able to rearrange their structure maintaining Fo. On the contrary, in archaea, which possess a more restricted metabolism (mainly hydrogenotrophic, acetoclastic and methylotrophic), change to a situation in which ethanol was the sole energy and carbon source, could have induced a strong variation in the substrates available for methane production by archaea from the inoculum sample to the first anaerobic condition (AN) in R1, leading to a community with a non-well-defined internal structure in terms of species dominance.

Oxygen introduction in R1 ($ASB_{0.1}$) led to a slight increase in Rr in bacterial communities, likely due to availability of an additional electron acceptor for new metabolic reactions (sulfide oxidation, ethanol oxidation). Fo remained at a medium value and H was similar, suggesting a low impact of O_2 introduction in terms of bacterial community structure. At this stage ($ASB_{0.1}$), archaeal communities were able to recover medium Fo values, suggesting the emergence of certain archaeal species as dominant, while remaining the vast majority in lower amounts. Rr still remained at low values (Table 4). The operation of R1 under microaerobic conditions at an airflow rate of 0.3 mL min^{-1} ($ASB_{0.3}$) promoted a negative impact on bacterial community, decreasing the ecological parameters (Fo from medium to low values, Rr from medium-high to low values, and H from 2.44 to 1.73). Archaeal communities presented similar values than those observed at stage $ASB_{0.1}$. Despite variations observed in richness and evenness of the bacterial and archaeal communities due to the operational conditions, R1 remained stable in terms of COD and sulfate removal when oxygen was applied at low concentrations. The highly dynamic communities (change in species coming to significance) likely ensured good process functionality as previously observed by other authors [38].

In R2, initially operated under microaerobic conditions at an airflow rate of 0.3mL min^{-1} ($HD_{0.3}$), a slight increase in diversity (H) and Rr (while still within the range of medium richness) of bacterial communities was observed likely due to the

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availability of O_2 as electron acceptor for bacteria (Table 4). Microaeration implies more ecological niches to be occupied by bacteria and thus an increase in diversity and richness can be observed. Fo remained at its medium values as compared to the inoculum sample, suggesting a low impact of feeding change on bacterial populations in terms of ecological structure. Similarly, a slight decrease in Rr was observed for archaea (while still within the range of high Rr values) and Fo remained at its medium values (Table 4).

The airflow reduction in R2 ($HD_{0.1}$) promoted a decrease in bacterial and archaeal Rr (from 86 (high) to 27 (medium) in bacteria; from 70 (high) to 26 (medium) in archaea) and H (from 3.05 to 2.40 in bacteria; form 2.47 to 1.98 in archaea), likely due to the low availability of O_2 for facultative and aerobic organisms (sulfide oxidizers) and to changes in the substrate type for methanogens, while Fo remained at a medium value (Table 4).

When microaeration was supplied at the sludge blanket ($SB_{0.1,}$ $SB_{0.3}$) major changes in Rr, Fo and H were not observed as compared to the previous operation stage ($HD_{0.1}$). A slight increase in diversity and Rr was observed for archaeal and bacterial communities but Fo remained at medium values (Table 4). Very similar values of Rr, H and Fo were observed between $SB_{0.1}$ and $SB_{0.3}$ stages, which corroborates that the disturbance applied to the system (increase in airflow rate) was not enough to induce shifts in the bacterial community.

The oxygen at 0.3 $mL \cdot min^{-1}$ affected mainly bacterial communities in R1, while ecological parameters of archaeal communities remained almost constant. The effects of increasing oxygen concentration consisted on a decrease in Fo and Rr parameters suggesting the loss of diversity. A plausible explanation could be related to an inhibitory effect of O_2 on sulfate reducing bacteria, syntrophic bacteria or both, leading to the loss of species within these groups. In addition, a decrease of these bacterial groups could be related to an increase of aerobic bacteria consuming the substrates previously used by these groups. In fact, significant differences in SO_4^{2-} removal were observed between $ASB_{0.1}$ and $ASB_{0.3}$ suggesting a negative effect of O_2 directly on SRB, over syntrophic communities providing the SRB growth substrate or caused by competition with aerobic microorganisms. Unlike in R1, using an airflow rate of 0.3 $mL \cdot min^{-1}$ did not affect bacterial-related ecological parameters in R2. The fact that R1 was firstly subjected to an anaerobic phase using ethanol as the sole substrate led to a decrease mainly in bacterial and archaeal richness due to reduction in resource variety.

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This specialization of anaerobic groups could have influenced the subsequent adaptation of these microorganisms when using and airflow rate of 0.3 $mL \cdot min^{-1}$ limiting the emergence of species able to tolerate such oxygen concentrations. On the contrary, R2 was not subjected to this specialization since it was directly operated under microaerobic conditions at an airflow rate of 0.3 $mL \cdot min^{-1}$. In fact, authors have reported the importance of the inoculum characteristics, mainly diversity, to stablish a good microbial community in terms of functionality [38].

Despite variations observed in ecological parameters measured in this study, a weak correlation was observed between these variations and the functional performance of the systems in terms of COD and sulfate removal, which remained almost stable along the whole reactor operation in R1 and R2. However, the higher levels of richness and diversity for archaea and bacteria were observed for the sample corresponding to $HD_{0.3}$, when the best sulfide removal capacity was recorded (Table 2), suggesting a higher availability of oxygen to sulfide oxidizers under this conditions.

4. Conclusions

The microaeration at the headspace represents a feasible and efficient strategy for sulfate and H_2S removal in low-strength wastewaters. The best airflow was 0.3 $mL \cdot min^{-1}$ ($HD_{0.3}$), resulting in the lowest H_2S (< 0.1%) concentration, whereas CH_4 concentration remained above 60%.

The highest productivity and operational efficiency was observed at this air dosing point, with this airflow ($HD_{0.3}$), which corroborates with the operational results and the ecological parameters, since the microaeration at this stage promoted high bacterial and archaeal species richness and diversity, optimum functional organization, high COD and H_2S removal efficiency.

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Disclosures

The authors indicate no potential conflicts of interest.

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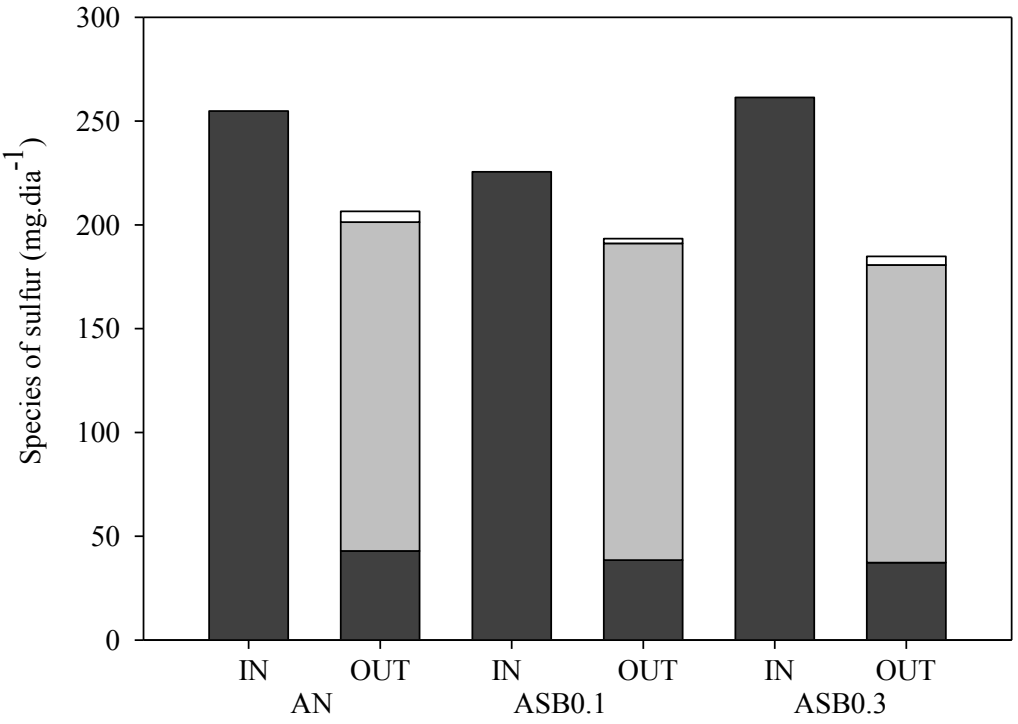
Figure captions

Fig. 1. Sulfur mass balance for the different air dosing strategies ($mg.d^{-1}$): sulfate (■); dissolved sulfide (▭) and H_2S (▭).

Fig. 2. Bacterial and archaeal DGGE profiles for the different experimental stages.

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Fig. 1


300
250
200
150
100
50
0
Species of sulfur (mg.dia-1)
IN
OUT
IN
OUT
IN
OUT
AN
ASB0.1
ASB0.3

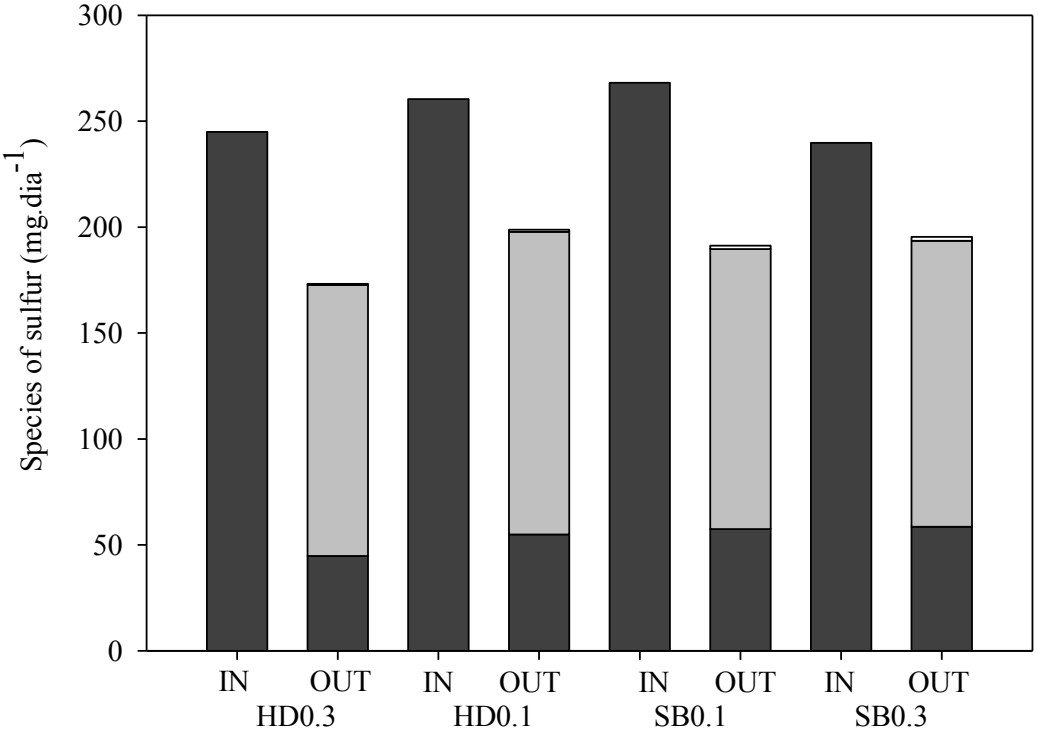
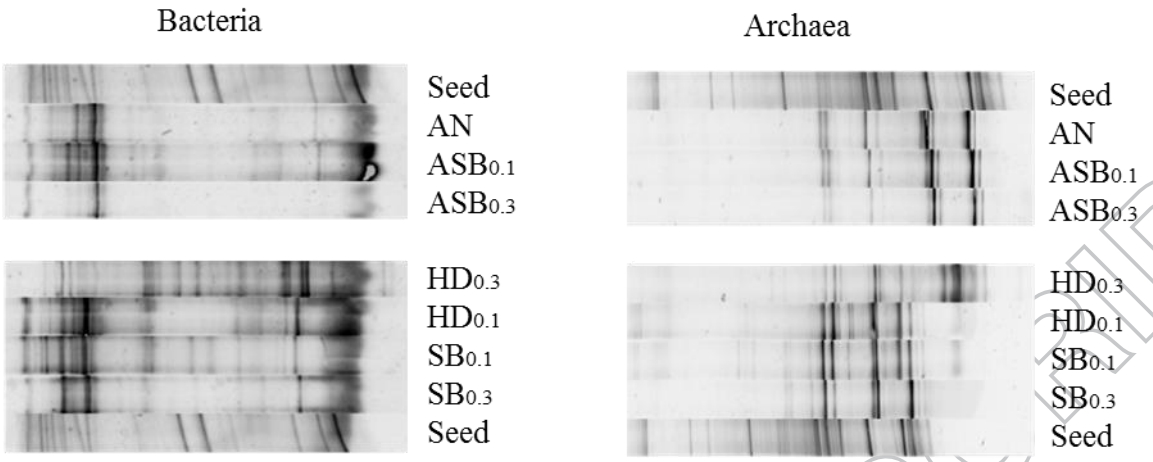

300
250
200
150
100
50
0
Species of sulfur (mg.dia-1)
IN
OUT
IN
OUT
IN
OUT
IN
OUT
HD0.3
HD0.1
SB0.1
SB0.3

Fig. 2



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Table 1. Operational parameters for the different air dosing strategies.

Parameter	**Unit**	**Reactors and stages**						
		Reactor 1				**Reactor 2**		
		AN	**$ASB_{0.1}$**	**$ASB_{0.3}$**	**$HD_{0.3}$**	**$HD_{0.1}$**	**$SB_{0.1}$**	**$SB_{0.3}$**
Microaeration dosing point	-	-	Above Sludge Blanket	Above Sludge Blanket	Headspace	Headspace	Sludge Blanket	Sludge Blanket
Airflow rate	$mL.min^{-1}$	-	0.1	0.3	0.3	0.1	0.1	0.3
Q_{air}/Q_{inf}	$L_{air}.L_{inf}^{-1}$		0.024	0.072	0.072	0.024	0.024	0.072
Days of operation	Day	155	90	64	37	43	63	53
HRT	H	11.0	11.0	11.0	11.0	11.0	11.0	11.0
Flow rate	$L \cdot d^{-1}$	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Influent COD	$mg.L^{-1}$	1000	1000	1000	1000	1000	1000	1000
OLR	$kgCOD \cdot m^{-3} \cdot day^{-1}$	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Influent SO_4^{2-}	$mg.L^{-1}$	140	140	140	140	140	140	140
SLR	$kgSO_4^{2-} \cdot m^{-3} \cdot day^{-1}$	0.3	0.3	0.3	0.3	0.3	0.3	0.3
COD/SO_4^{2-}		7.1	7.1	7.1	7.1	7.1	7.1	7.1

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Table 2. Operational performance for the different air dosing strategies.

Parameter	Unit	Reactor and stages						
		Reactor 1				Reactor 2		
		AN	**$ASB_{0.1}$**	**$ASB_{0.3}$**	**$HD_{0.3}$**	**$HD_{0.1}$**	**$SB_{0.1}$**	**$SB_{0.3}$**
Microaeration dosing point	-	-	Above Sludge Blanket	Above Sludge Blanket	Headspace	Headspace	Sludge Blanket	Sludge Blanket
Airflow rate	$mL.min^{-1}$	-	0.1	0.3	0.3	0.1	0.1	0.3
Q_{air}/Q_{inf}	$L_{air}.L_{inf}^{-1}$		0.024	0.072	0.072	0.024	0.024	0.072
Days of operation	Day	155	90	64	37	43	63	53
Effluent COD	$mg.L^{-1}$	199 (206)	74 (44)	105 (96)	225 (44)	127 (65)	123 (556)	122 (37)
COD removal	%	80 (23)	92 (5)	87 (13)	89 (10)	84 (4)	79 (13)	84 (5)
Influent SO_4^{2-}	$mg.L^{-1}$	139 (17)	140 (15)	123 (29)	134 (19)	135 (22)	139 (17)	124 (29)
Effluent SO_4^{2-}	$mg.L^{-1}$	23 (18)	20 (7)	21 (5)	24 (15)	28 (5)	30 (31)	30 (22)
SO_4^{2-} removal	%	83 (11)	86 (6)	77 (11)	81 (12)	78 (6)	79 (22)	68 (17)
Dissolved S^{2-}	$mg.L^{-1}$	30 (8)	29 (10)	26 (6)	24 (10)	25 (10)	24 (13)	24 (13)
H_2S	%	0.50 (0.40)	0.34 (0.20)	0.17 (0.10)	0.03 (0.04)	0.07 (0.08)	0.10 (0.02)	0.13 (0.04)
	$mmol.d^{-1}$	0.24 (0.20)	0.16 (0.09)	0.08 (0.05)	0.01 (0.02)	0.03 (0.04)	0.05 (0.01)	0.07 (0.03)

The standard deviation is shown in parentheses.

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Table 3. Average values of concentration (%) and molar flow rate (mmol.d^{-1}) for the gases O_2, N_2, CH_4, CO_2, H_2S and NH_3.

Parameter	**Unit**	**Reactor and stages**						
		Reactor 1				**Reactor 2**		
		AN	**$ASB_{0.1}$**	**$ASB_{0.3}$**	**$HD_{0.3}$**	**$HD_{0.1}$**	**$SB_{0.1}$**	**$SB_{0.3}$**
Microaeration dosing point	-	-	Above Sludge Blanket	Above Sludge Blanket	Headspace	Headspace	Sludge Blanket	Sludge Blanket
Airflow ratr	mL.min^{-1}	-	0.1	0.3	0.3	0.1	0.1	0.3
Q_{air}/Q_{inf}	$L_{air}.L_{inf}^{-1}$		0.024	0.072	0.072	0.024	0.024	0.072
O_2	%	0.16 (0.08)	0.11 (0.03)	0.20 (0.01)	0.87 (0.17)	0.69 (0.60)	0.39 (0.60)	1.73 (0.54)
	mmol.d^{-1}	0.08 (0.04)	0.05 (0.02)	0.10 (0.01)	0.42 (0.10)	0.34 (0.30)	0.19 (0.30)	0.82 (0.20)
N_2	%	13.1 (7.0)	19.6 (13.6)	25.5 (19.6)	36.4 (14.6)	21.6 (11.9)	17.5 (4.4)	14.6 (2.6)
	mmol.d^{-1}	6.4 (3.4)	9.6 (6.6)	14.1 (11.3)	17.7 (7.1)	10.5 (5.8)	8.5 (2.1)	9.7 (6.4)
CH_4	%	80.0 (6.6)	75.0 (13.2)	69.3 (18.6)	62.6 (10)	71.8 (11.3)	75.7 (4.3)	77.2 (2.2)
	mmol.d^{-1}	39.0 (3.2)	36.6 (6.5)	32.4 (10.7)	28.0 (7.1)	34.9 (5.5)	36.9 (2.1)	34.6 (7.9)
CO_2	%	5.4 (1.6)	3.9 (4.6)	4.1 (4.7)	5.2 (0.7)	5.6 (0.8)	6.0 (0.5)	6.0 (0.4)
	mmol.d^{-1}	2.6 (0.8)	1.9 (2.2)	1.9 (2.0)	2.5 (0.0)	2.7 (0.4)	3.0 (0.2)	3.3 (2.3)
H_2S	%	0.50 (0.40)	0.34 (0.20)	0.17 (0.10)	0.03 (0.04)	0.07 (0.08)	0.10 (0.02)	0.14 (0.04)
	mmol.d^{-1}	0.24 (0.20)	0.16 (0.09)	0.08 (0.05)	0.01 (0.02)	0.03 (0.04)	0.05 (0.01)	0.07 (0.03)
NH_3	%	0.83 (0.40)	0.92 (0.40)	0.61 (0.40)	0.05 (0.06)	0.18 (0.20)	0.31 (0.09)	0.47 (0.20)
	mmol.d^{-1}	0.40 (0.20)	0.45 (0.20)	0.30 (0.10)	0.02 (0.03)	0.09 (0.10)	0.15 (0.04)	0.12 (0.10)

The standard deviation is shown in parentheses.

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Table 4. Functional organization (Fo), Range-weighted richness (Rr) and Shannon diversity index (H) obtained from the DGGE profiles.

Parameter	Inoculum	Reactor 1			Reactor 2			
		AN	$ASB_{0.1}$	$ASB_{0.3}$	$HD_{0.3}$	$HD_{0.1}$	$SB_{0.1}$	$SB_{0.3}$
Bacteria								
Functional Organization (Fo) (%)	33	38	53	21	43	49	45	40
Range-Weighted Richness (Rr)	50	17	33	5	86	27	36	36
Shannon Diversity (H)	2.77	2.31	2.44	1.73	3.05	2.40	2.68	2.77
Archaea								
Functional Organization (Fo) (%)	45	8	35	39	41	55	46	55
Range-Weighted Richness (Rr)	193	9	9	6	70	26	45	63
Shannon Diversity (H)	3.18	2.05	2.04	1.74	2.47	1.98	2.31	2.35

Rr: low (<10), average (10-30), high (>30) according to [17]
Fo: low Fo/high evenness (25%), average Fo/evenness (30-70%), high Fo/low evenness (70%) according to [17]
H: 1.5 to 3.5 (low evenness and high richness) according to [39]

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